Project 02
AI Music & Audio Analyzer

Test & Development Report — v0.7

Item	Current Value
Status	In Progress — hardware, DSP, real music analysis and visualization verified
Platform	Windows Laptop
Python	3.14.7
Environment	VS Code + PowerShell + Python venv
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Music Input	test_music.mp3
Libraries	sounddevice + Librosa + NumPy + Matplotlib
Current Milestone	Step 14 — Professional Music Analysis Dashboard

1. Project Objective

Build a practical Python-based AI Music & Audio Analyzer using a Windows laptop and Airdopes Joy Bluetooth earbuds. The project progressively validates Bluetooth audio capture, digital signal processing, spectral analysis, frequency-band analysis, rhythm analysis, real music input and professional visualization.

2. Completed Steps

Step	Feature	Status
1	Audio Device Detection	PASS
2	Airdopes Microphone Verification	PASS
3	Airdopes WAV Recording	PASS
4	WAV Audio Analysis	PASS
5	Waveform Visualization	PASS
6	Audio Level Analysis	PASS
7	FFT / Frequency Spectrum	PASS
8	STFT / Spectrogram	PASS
9	Spectral Feature Analysis	PASS
10	Bass / Mid / Treble Energy	PASS
11	Tempo / Beat Analysis	PASS
12	Integrated Audio Feature Summary	PASS
13	Real Music File Input & Analysis	PASS
14	Professional Music Visualization Dashboard	PASS

3. Airdopes Hardware Test

Parameter	Result
Device ID	2
Device	Airdopes Joy Hands-Free
Input	1 channel, 44,100 Hz
Duration	5.00 sec
File	audio\airdopes_test.wav
RMS Level	-30.15 dBFS
Peak Level	-7.45 dBFS
Crest Factor	13.638

4. Airdopes Signal Analysis

Feature	Result
Dominant Frequency	544.60 Hz
Spectral Centroid	7,159.79 Hz
Bandwidth	4,349.76 Hz
Rolloff	12,150.42 Hz
Flatness	0.329063
Zero Crossing Rate	0.147712
Brightness	Bright
Texture	Noise-like
Bass / Mid / Treble	3.49% / 96.48% / 0.03%
Estimated BPM	147.66
Beats / Onsets	6 / 15

The Airdopes recording is a short microphone test, so its BPM and music classifications are pipeline-validation results rather than definitive musical properties.

5. Real Music Track — Step 13

Parameter	Actual Result
File	test_music.mp3
Duration	240.02 sec
Sample Rate	44,100 Hz
Channels	1 (mono)
Samples	10,585,088
RMS Level	-17.05 dBFS
Peak Level	-0.17 dBFS
Spectral Centroid	2,315.86 Hz
Bandwidth	3,010.12 Hz
Rolloff	5,069.88 Hz
Flatness	0.000201
ZCR	0.046885
Bass / Mid / Treble	89.42% / 10.50% / 0.08%
Dominant Band	Bass
Estimated BPM	109.96

Parameter	Actual Result
Beats / Onsets	437 / 1,052
Tempo Category	Moderate

6. Airdopes vs Music Comparison

Feature	Airdopes Test	Music Track
Duration	5.00 sec	240.02 sec
RMS Level	-30.15 dBFS	-17.05 dBFS
Peak Level	-7.45 dBFS	-0.17 dBFS
Centroid	7,159.79 Hz	2,315.86 Hz
Flatness	0.329063	0.000201
Bass	3.49%	89.42%
Mid	96.48%	10.50%
Treble	0.03%	0.08%
BPM	147.66	109.96
Beats	6	437
Onsets	15	1,052

7. Integrated Analyzer — Step 12

Step 12 combines signal level, FFT, spectral features, Bass/Mid/Treble analysis and tempo/beat analysis into one unified terminal summary, establishing the core analyzer engine.

Integrated Feature	Result
Signal Level	Healthy
RMS	-30.15 dBFS
Peak	-7.45 dBFS
Crest Factor	13.638
Dominant Frequency	544.60 Hz
Dominant Band	Mid
Brightness	Bright
Texture	Noise-like
BPM	147.66
Status	COMPLETE

8. Professional Visualization Dashboard — Step 14

The first dashboard was functional but visually dense. It was refined into a clearer professional v2 dashboard with KPI cards, full-track waveform, logarithmic frequency spectrum, frequency-balance bars, focused spectrogram and a dedicated spectral/rhythm information panel.

Dashboard Element	Purpose
KPI Cards	BPM, RMS dBFS, Peak dBFS, dominant band
Waveform	Full 240-second track overview
Frequency Spectrum	Log-frequency view for easier music interpretation
Frequency Balance	Bass/Mid/Treble relative-energy bars
Spectrogram	Time-frequency visualization focused to 10 kHz
Feature Panel	Centroid, bandwidth, rolloff, flatness, ZCR, BPM, beats, levels
Final Output	reports\music_analysis_dashboard_v2.png

9. Current Project Structure

Project-02-AI-Music-Audio-Analyzer/

■■■ audio/ → airdopes_test.wav

■■■ music/ → test_music.mp3

■■■ reports/ → Airdopes analysis graphs + music_analysis_dashboard.png + music_analysis_dashboard_v2.png

■■■ src/ → device detection, recording, WAV analysis, waveform, level, FFT, spectrogram, spectral features, Bass/Mid/Treble, tempo/beat, integrated summary, music analysis and dashboard scripts

■■■ tests/

■■■ venv/

■■■ requirements.txt

■■■ README.md

10. Overall Validation

Validation	Status
Airdopes Bluetooth microphone accessible	PASS
Airdopes recording	PASS
WAV signal analysis	PASS
Waveform / FFT / spectrogram	PASS
Spectral features	PASS
Bass/Mid/Treble	PASS
Tempo/beat analysis	PASS
Integrated analyzer	PASS
Real music input	PASS
Real music analysis	PASS
Professional dashboard	PASS
AI interpretation	PENDING

11. Next Development Step

Step 15 — Advanced Music Features: key/chroma analysis, loudness and dynamic-range analysis, beat-synchronous features and, later, music mood or genre estimation. These should be validated against the real music track.

12. Version History

v0.1 — New Project-02 documentation baseline.

v0.2 — Steps 4–6: WAV analysis, waveform and audio-level analysis.

v0.3 — Steps 7–8: FFT spectrum and spectrogram.

v0.4 — Step 9: spectral features.

v0.5 — Steps 10–11: Bass/Mid/Treble and Tempo/Beat.

v0.6 — Steps 12–13: integrated summary and real music analysis.

v0.7 — Step 14: professional music visualization dashboard and refined v2 layout.

Only Project-02 development history is included.